LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve, Lo
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve, Longitudinal dynamic stability,
5. Specific Objectives	11.1.4.1 Trainee will be able to describe aircraft stability in all the three axis

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of longitudinal stability concepts using IB diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define aircraft stability in the three axes.
(b) List the primary forces acting on a helicopter in longitudinal flight.
(c) Explain the difference between stick fixed and stick free stability.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces and moments in longitudinal axis • Stick fixed and stick free stability • Neutral point and static margin • Stability curve characteristics • Longitudinal dynamic stability	L & Ds	PPT, IB	15 min
(b)	• Analysis of stability curve behavior • Calculation of static margin • Dynamic stability oscillation modes • Comparison of stick fixed and free • Impact of CG location on stability	L & Ds	PPT, PDF	20 min
(c)	• Summary of longitudinal stability principles • Review of neutral point significance • Application of stability concepts • Preparation for lateral stability • Q&A session on dynamic stability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the relationship between the neutral point and the center of gravity.
(b) Differentiate between static and dynamic longitudinal stability.
(c) Explain how the stability curve indicates aircraft stability characteristics.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from longitudinal to lateral stability focusing on the forces and moments involved, referencing the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Lateral stability: forces and Moments, effect of wing dihedral, effects of slide slip, induced roll and induced yaw. Factors effecting lateral stability. Wing sweepback effect. Dihedral effect	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min